

Kinetic segregation

Physics 230A/146A — Biological Physics

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This deck is a **placeholder**. It exists so the slide pipeline is exercised rather than asserted — every construct below is here to be checked in all three renders, not because it is the lecture. The real Lesson 5 deck replaces it.

What the model is for

Two cells, pressed together

A T cell touches an antigen-presenting cell, and the two membranes are held some distance apart.

Two things set that distance, and they pull against each other:

- ▶ **Binding** wants the receptor and its ligand at their natural span
- ▶ **Mechanics** wants the membrane flat

The contact is a competition between binding energy and bending energy.

The separation that costs least

The membrane pays to bend, and a bond pays to stretch. Adding the two gives an energy with a minimum at some separation **h-nought**.

A figure of the geometry belongs here on the page. The deck carries a different one, sized for a projector; this pair is what `.slide-only` and `.page-only` are for, and `linear.lua` switches between them.

A first pass at the energy

```
h = np.linspace(0.2, 3.0, 400) # separation, in units of the bond's natural span

bend = 0.5 * h**2                # membrane bending, growing with separation
bond = -2.0 * np.exp(-((h - 1.0) ** 2) / 0.1) # a binding well at h = 1

fig, ax = plt.subplots()
ax.plot(h, bend, "--", label="bending")
ax.plot(h, bond, "--", label="binding")
ax.plot(h, bend + bond, "-k", lw=2, label="total")
ax.set_xlabel("Separation $h$")
ax.set_ylabel("Energy")
ax.legend()
plt.show()
```

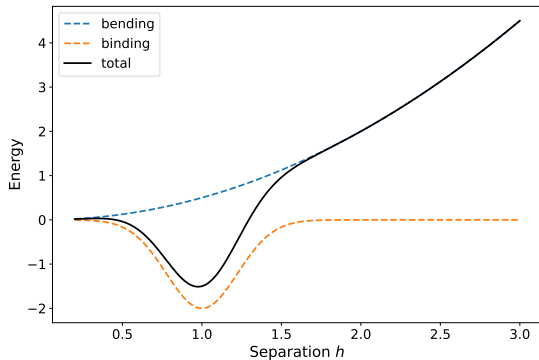


Figure 1: A bending cost rising with separation against a binding well, and their sum.

What to take from it

Holds up

The competition picture gives a single preferred separation without anything being tuned.

The second half of the lesson — diffusion of the large molecules out of the contact zone — has no slides yet. That is the part PS5 actually asks about.

Does not

Nothing here is diffusion yet, and the real question is what happens when a *second*, longer molecule is in the contact.